

Practice Problems:
Useful Techniques of Circuit Analysis
Introduction to Electric Circuits
Supplemental Instruction

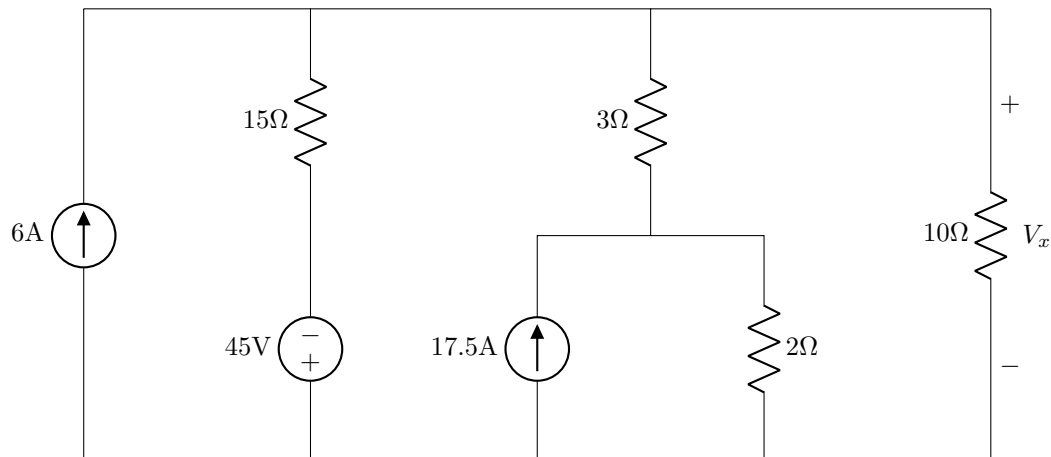
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March 26 2026

This document is not a substitute for a textbook or classroom instruction. It may contain errors and is intended to be used only to the extent that it is helpful to supplement students' learning experience.

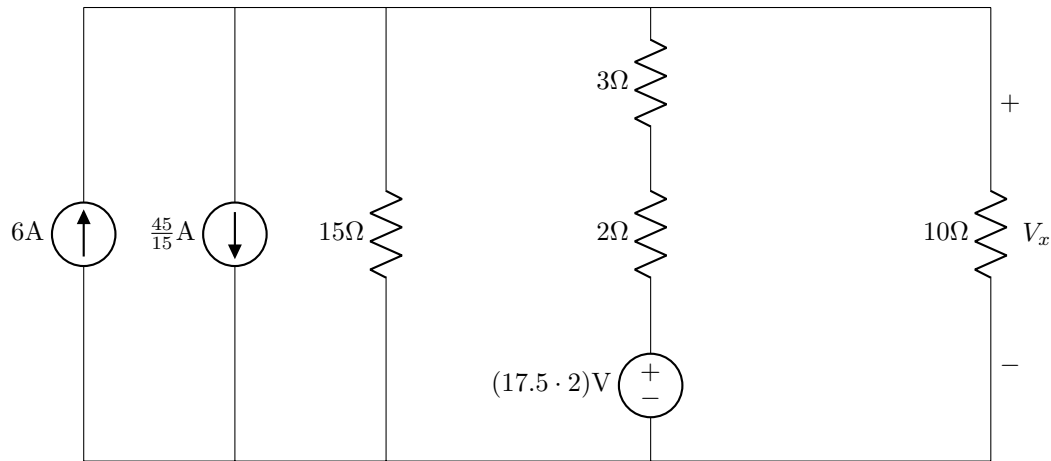
1 Source Conversions

Find V_x .

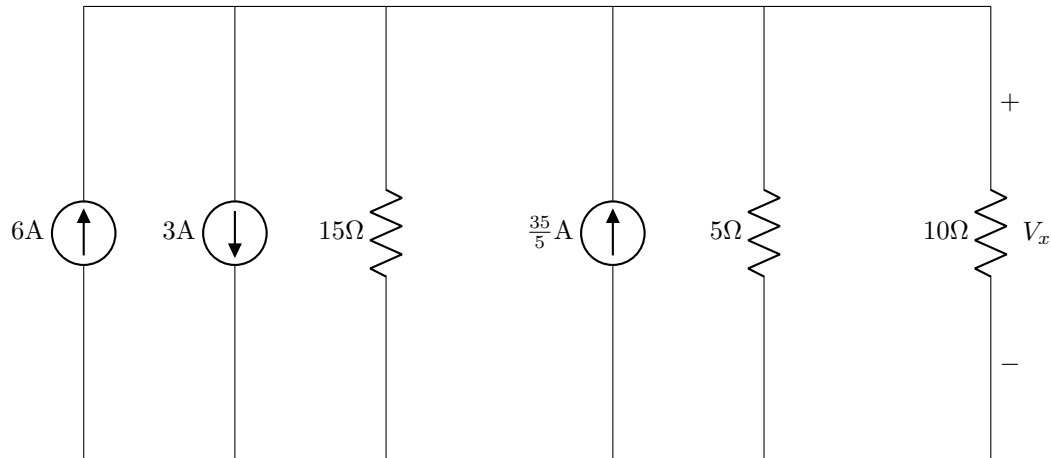


Solution

We will apply a source conversion to the voltage source in series with the 15Ω resistor, and apply a source conversion to the 17.5A current source in parallel with the 2Ω resistor.



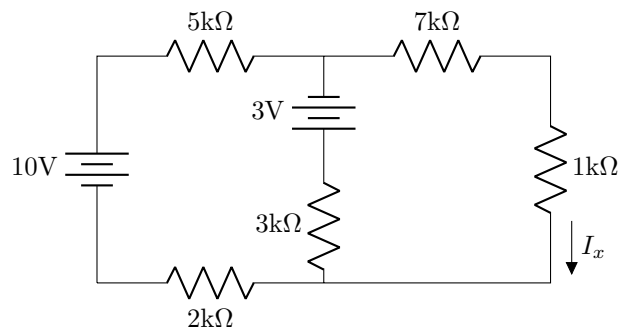
Now we will apply a source conversion to the voltage source in series with a combined 5Ω resistance.



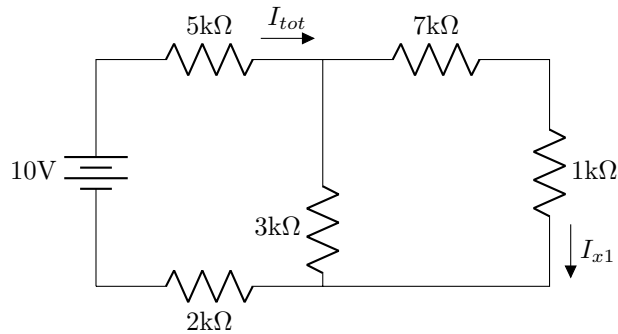
The current through the 10Ω resistor can be found using current division and KCL: $I_x = (6A + 7A - 3A) \cdot \frac{\frac{1}{10}}{\frac{1}{10} + \frac{1}{15} + \frac{1}{5}} = 2.727A$. Now the voltage can be found with Ohm's Law: $V_x = 2.727A \cdot 10\Omega = 27.27V$.

2 Superposition

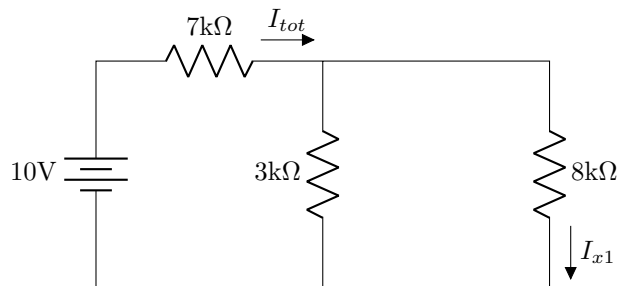
Find I_x .



We will first turn off the 3 V source:

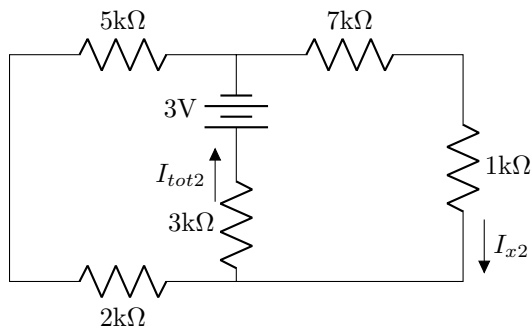


We can combine resistors to obtain:

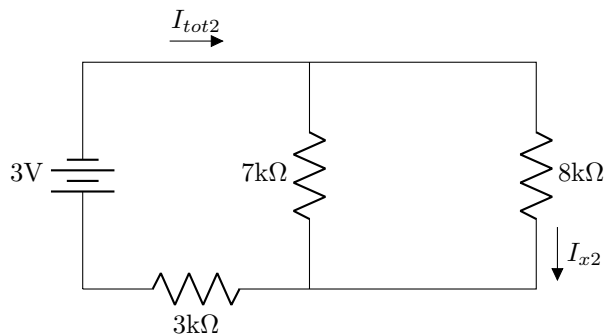


The two resistors on the right can be combined in parallel: $(3\text{k}\Omega || 8\text{k}\Omega) = 2.18182\text{k}\Omega$. We can find the total current $I_{tot} = \frac{10\text{V}}{7\text{k}\Omega + 2.18182\text{k}\Omega} = 1.089\text{ mA}$. Now, using current division, we can find $I_{x1} = 1.089\text{mA} \cdot \frac{\frac{1}{8}}{\frac{1}{8} + \frac{1}{3}} = 297\text{ }\mu\text{A}$.

Next, we will turn off the 10 V source:



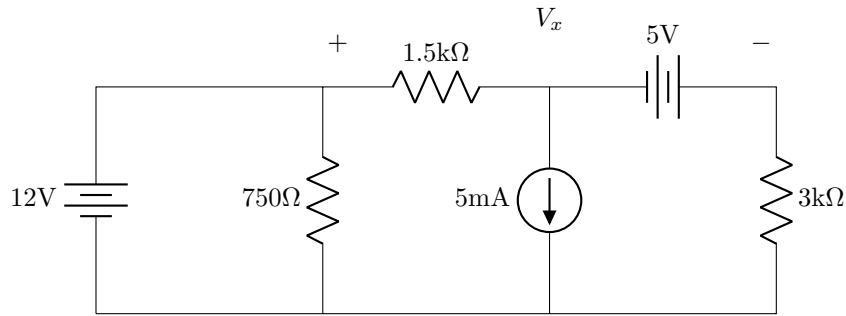
Combining resistors and rearranging:



The two resistors on the right can be combined in parallel: $(7\text{k}\Omega \parallel 8\text{k}\Omega) = 3.7333\text{k}\Omega$. We can find the total current $I_{tot2} = -\frac{3\text{V}}{3\text{k}\Omega + 3.7333\text{k}\Omega} = -446\text{ }\mu\text{A}$. Now, using current division, $I_{x2} = -446\text{ }\mu\text{A} \cdot \frac{\frac{1}{8}}{\frac{1}{8} + \frac{1}{7}} = -208\text{ }\mu\text{A}$. So, combining the two cases, $I_x = I_{x1} + I_{x2} = 297\text{ }\mu\text{A} + (-208\text{ }\mu\text{A}) = 89\text{ }\mu\text{A}$.

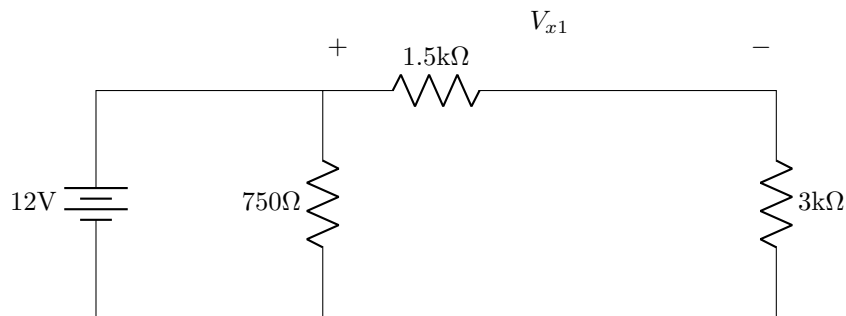
3 Superposition

Find V_x .



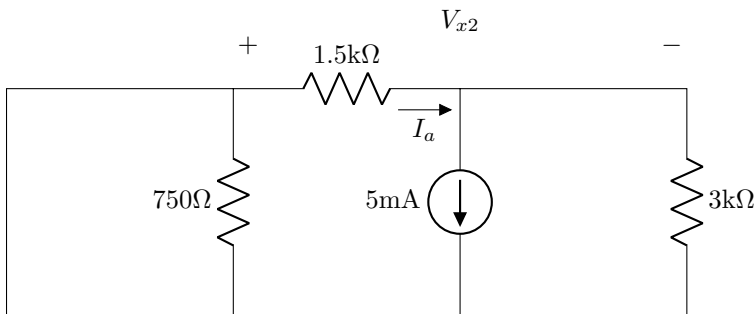
Solution

First, we leave the 12 V source on and turn off the other sources:



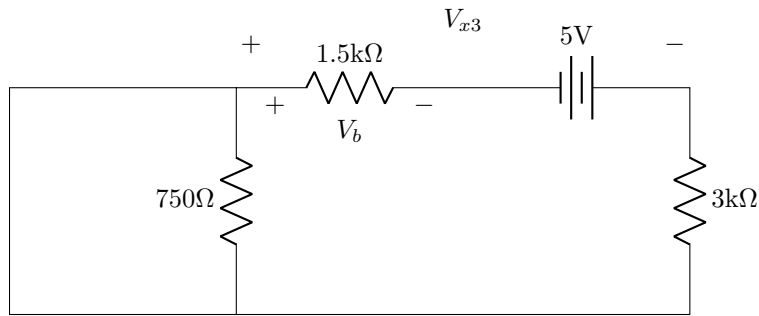
Using voltage division, $V_{x1} = 12\text{V} \cdot \frac{1.5}{1.5+3} = 4\text{ V}$.

Next, we leave the current source on and turn off the other sources:



The 750Ω resistor is shorted (no current flows through it). Using current division, $I_a = 5\text{mA} \cdot \frac{\frac{1}{1.5}}{\frac{1}{1.5} + \frac{1}{3}} = 3.33\text{mA}$. Using Ohm's Law, $V_{x2} = 3.33\text{mA} \cdot 1.5\text{k}\Omega = 5\text{ V}$.

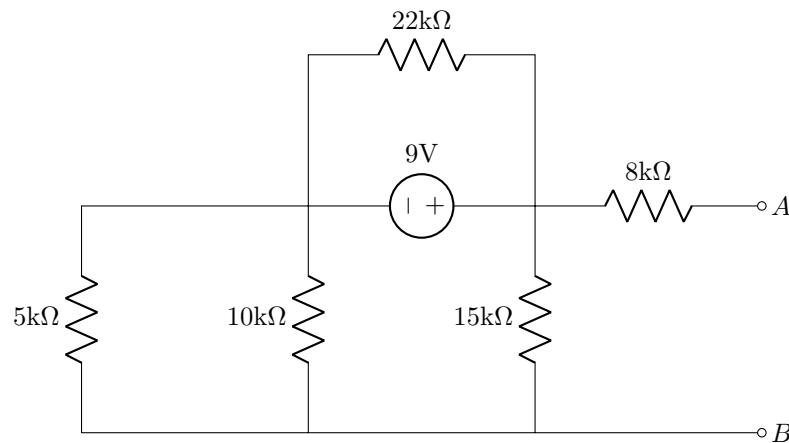
Next, we leave the 5 V source on and turn off the other sources:



The 750Ω resistor is shorted (there is no voltage across it). Using voltage division, $V_b = 5V \cdot \frac{1.5}{1.5+3} = 1.667V$. Using KVL, $V_{x3} = 1.667 - 5 = -3.33$ V. So, combining all three cases, $V_x = 4V + 5V - 3.33V = 5.67$ V.

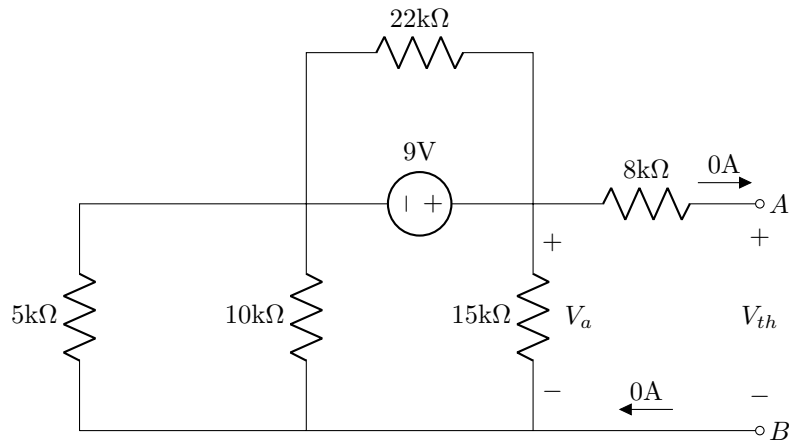
4 Thevenin Equivalent

Find the thevenin equivalent between points A and B.

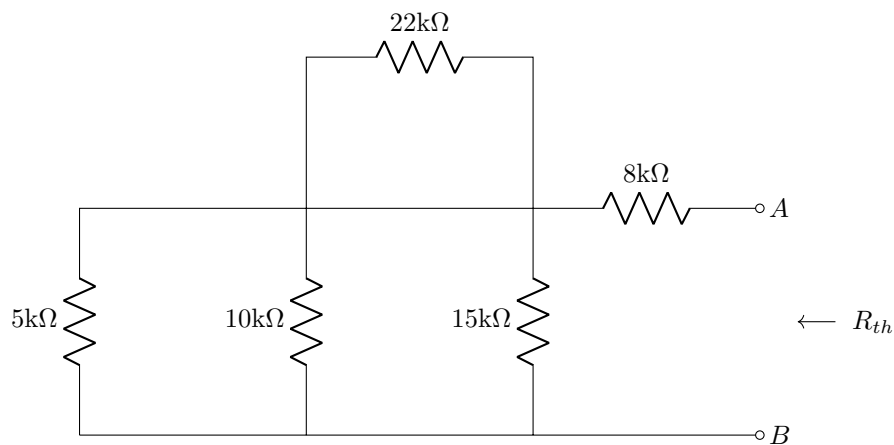


Solution

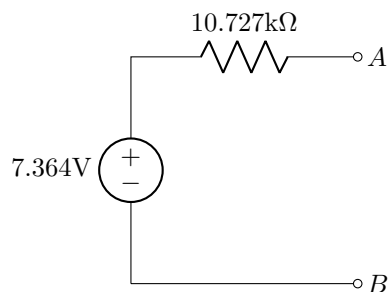
The thevenin voltage is found as the open-circuit voltage, that is, when there is no current flowing into or out of the network:



Since the current through the $8\text{k}\Omega$ resistor is 0, the voltage generated across it is 0, and $V_a = V_{th}$ by KVL. Also, note that the $15\text{k}\Omega$ resistor is in series with the two resistors on the left. We can find this voltage using voltage division: $V_{th} = V_a = 9\text{V} \cdot \frac{15\text{k}}{15\text{k} + (5\text{k} || 10\text{k})} = 7.364\text{ V}$. Next, the thevenin resistance is found with the source set to 0:

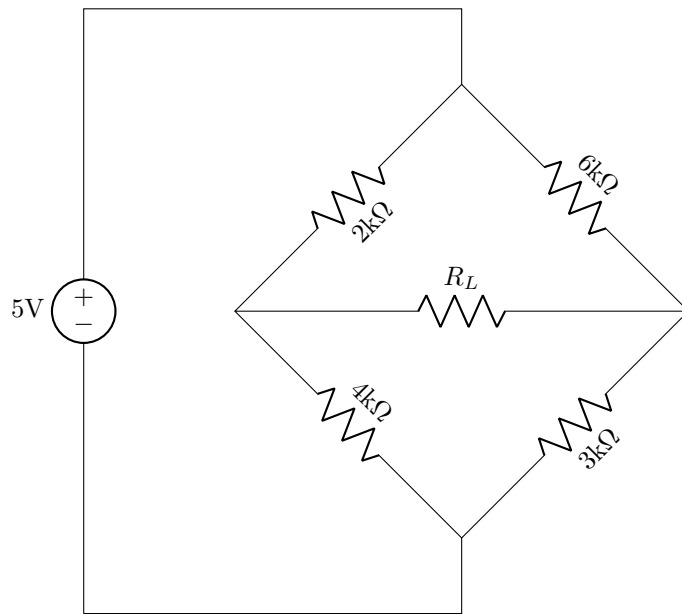


Note that the $22\text{k}\Omega$ resistor is shorted. $R_{th} = 8\text{k} + 5\text{k} || 10\text{k} || 15\text{k} = 10.727\text{ k}\Omega$. Therefore, the thevenin equivalent is:



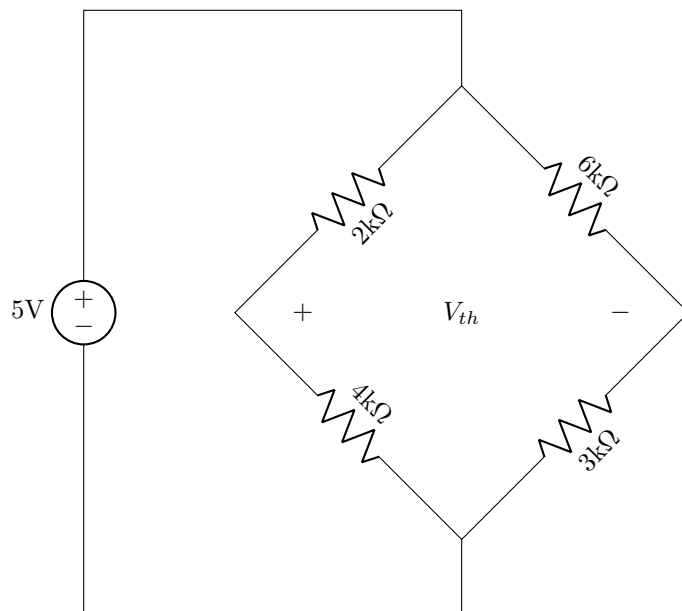
5 Wheatstone Bridge

Find the value of R_L required for a current of $200\ \mu\text{A}$ through R_L .

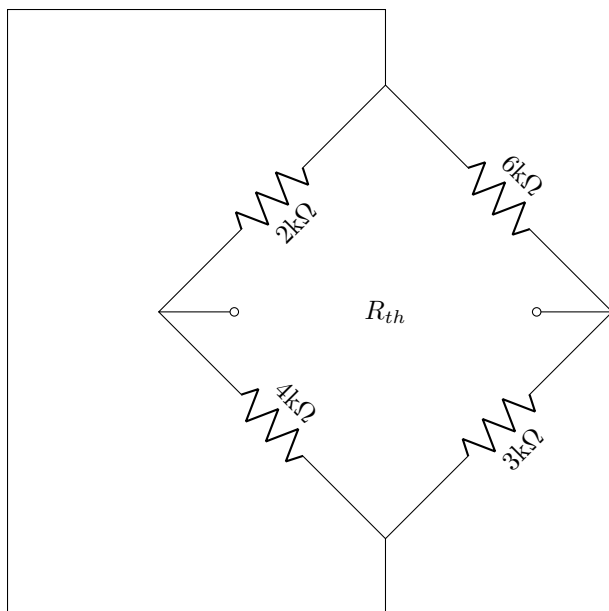


Solution

Let's find the thevenin equivalent of the bridge with R_L removed.

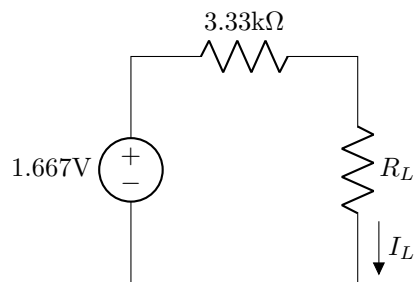


Using voltage division, $V_{th} = 5 \cdot \frac{4}{4+2} - 5 \cdot \frac{3}{3+6} = 1.667\ \text{V}$.
The thevenin resistance is found with the source turned off:



Note that the two resistors on the left are in parallel, and the two resistors on the right are in parallel. Therefore, $R_{th} = (2k||4k) + (6k||3k) = 3.33 \text{ k}\Omega$.

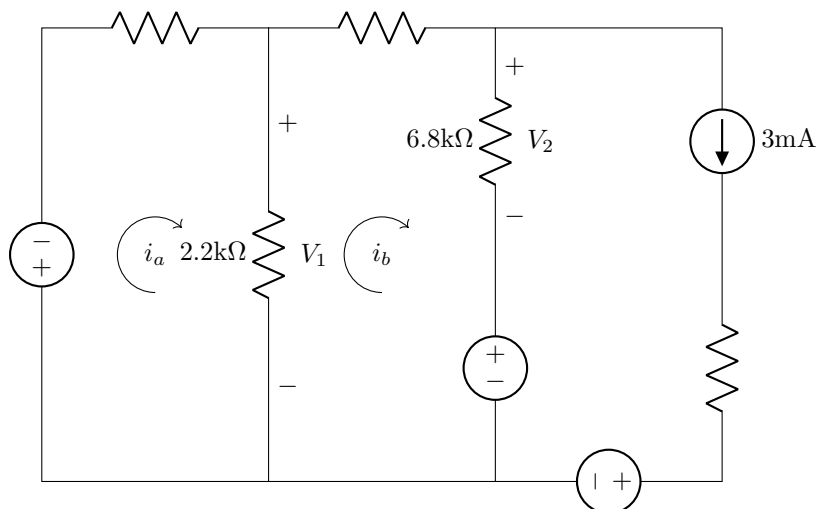
We can now use the thevenin equivalent to redraw the circuit more simply:



We can now find R_L : $I_L = 200\mu\text{A} = \frac{1.667\text{V}}{3.33\text{k}\Omega + R_L} \Rightarrow R_L = 5 \text{ k}\Omega$.

6 Mesh Currents

Find V_1 and V_2 given the mesh currents $i_a = 1\text{mA}$ and $i_b = 5\text{mA}$.

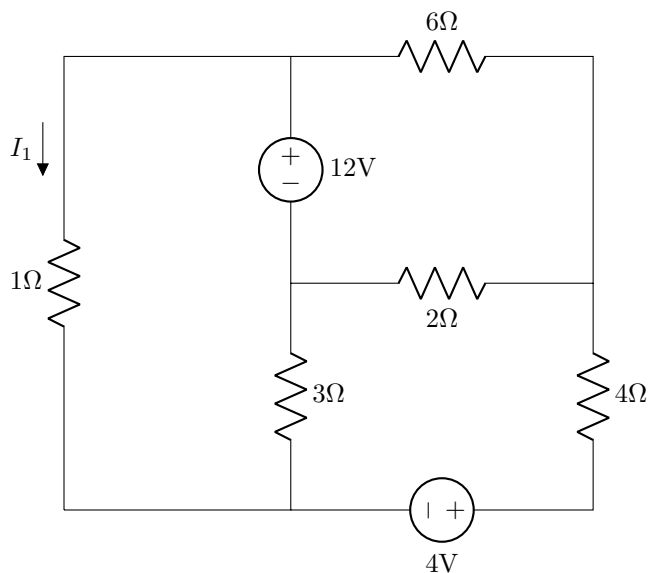


Solution

Note that the current of the rightmost mesh is given by the current source. The voltages can be found directly using KCL and Ohm's Law as in Mesh Analysis: $V_1 = 2.2\text{k}\Omega \cdot (i_a - i_b) = 2.2\text{k}\Omega \cdot (1\text{mA} - 5\text{mA}) = -8.8\text{ V}$, and $V_2 = 6.8\text{k}\Omega \cdot (i_b - 3\text{mA}) = 6.8\text{k}\Omega \cdot (5\text{mA} - 3\text{mA}) = 13.6\text{ V}$.

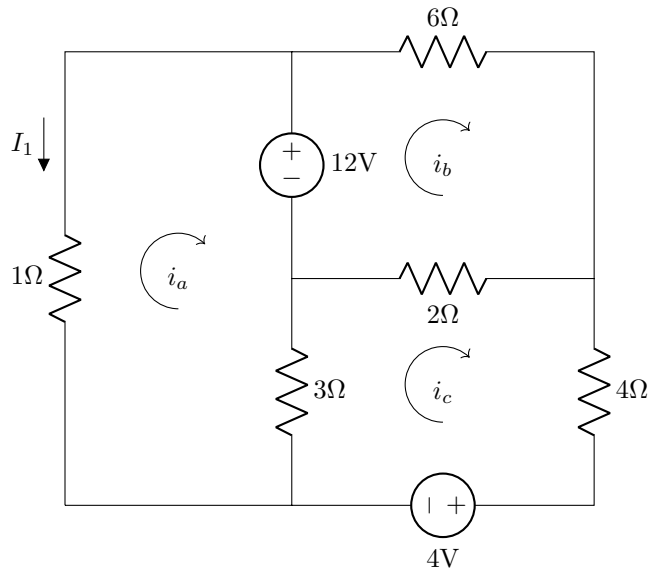
7 Mesh Analysis

Find I_1 .



Solution

Let's draw the three mesh currents flowing clockwise:



Now we can write the KVL equations for each mesh. We'll start at the upper left hand corner of each mesh and sum the voltages going clockwise:

For the mesh i_a :

$$12V + 3\Omega(i_a - i_c) + 1\Omega(i_a) = 0$$

For the mesh i_b :

$$6\Omega(i_b) + 2\Omega(i_b - i_c) - 12V = 0$$

For the mesh i_c :

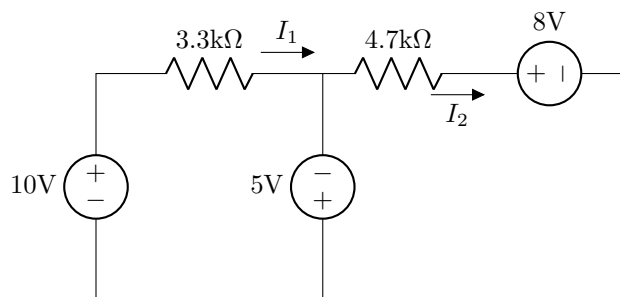
$$2\Omega(i_c - i_b) + 4\Omega(i_c) + 4V + 3\Omega(i_c - i_a) = 0$$

Now we can solve the system of equations using any method we like to obtain $i_a = -4.2$ A, $i_b = 1.1$ A, and $i_c = -1.6$ A.

We can now find $I_1 = -i_a = 4.2$ A.

8 Nodal Analysis Currents

Find I_1 and I_2 using nodal analysis techniques.

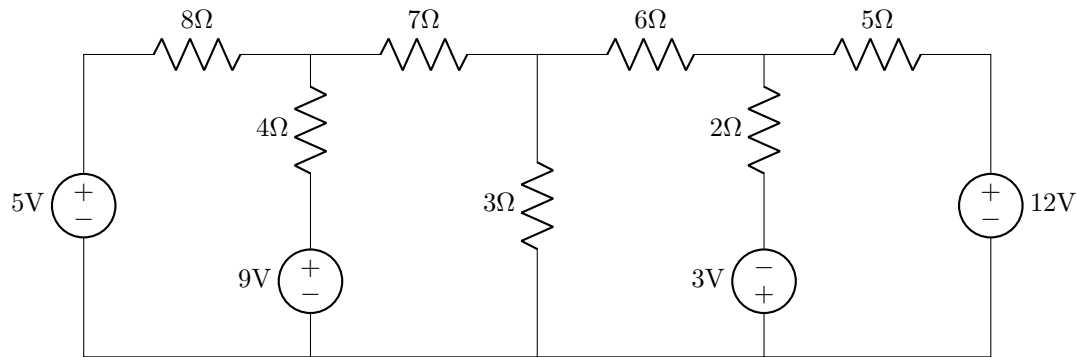


Solution

The currents can be found directly using KVL and Ohm's Law as in nodal analysis: $I_1 = \frac{10\text{V} - (-5\text{V})}{3.3\text{k}\Omega} = 4.545 \text{ mA}$, and $I_2 = \frac{(-5\text{V}) - 8\text{V}}{4.7\text{k}\Omega} = -2.766 \text{ mA}$.

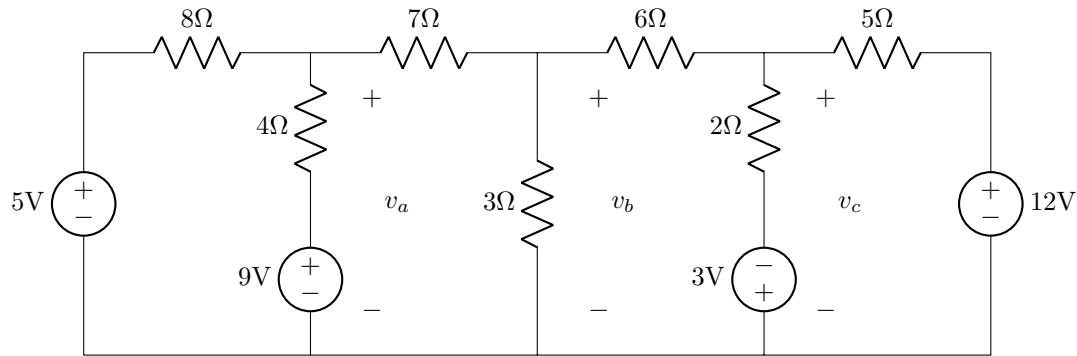
9 Nodal Analysis

Find the unknown nodal voltages with respect to the bottom node.



Solution

Let's name and label the nodal voltages:



Now we can write the nodal KCL equations:

For the node of voltage v_a :

$$\frac{5\text{V} - v_a}{8\Omega} + \frac{v_b - v_a}{7\Omega} + \frac{9\text{V} - v_a}{4\Omega} = 0$$

For the node of voltage v_b :

$$\frac{v_a - v_b}{7\Omega} + \frac{v_c - v_b}{6\Omega} - \frac{v_b}{3\Omega} = 0$$

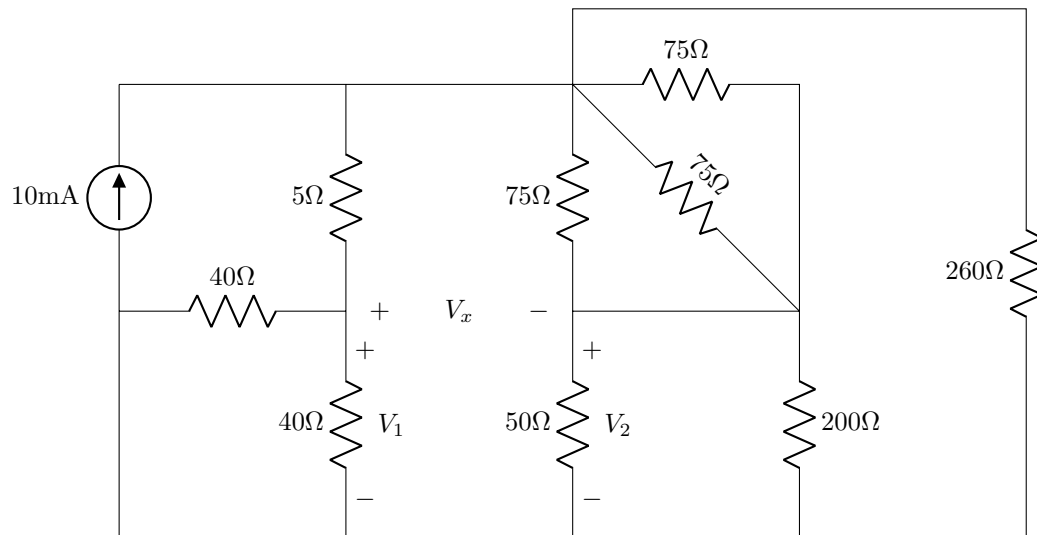
For the node of voltage v_c :

$$\frac{v_b - v_c}{6\Omega} + \frac{12\text{V} - v_c}{5\Omega} + \frac{(-3\text{V}) - v_c}{2\Omega} = 0$$

Solving these equations gives $v_a = 6.018\text{V}$, $v_b = 1.691\text{V}$, and $v_c = 1.364\text{V}$.

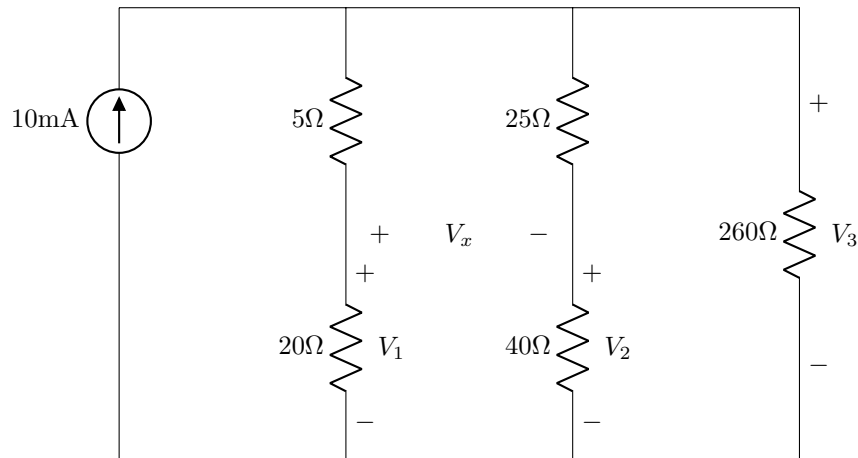
10 Nodal Analysis

Find V_x using nodal analysis.



Solution

To make the analysis easier, we can first simplify parallel resistors:



Setting the bottom node as the reference, we can now write the following KCL equations for nodes of voltage V_1 , V_2 , and V_3 :

$$\begin{aligned}\frac{V_3 - V_1}{5\Omega} + \frac{0 - V_1}{20\Omega} &= 0 \\ \frac{V_3 - V_2}{25\Omega} + \frac{0 - V_2}{40\Omega} &= 0 \\ 10\text{mA} + \frac{V_1 - V_3}{5\Omega} + \frac{V_2 - V_3}{25\Omega} + \frac{0 - V_3}{260\Omega} &= 0\end{aligned}$$

Solving, $V_1 = 135.065\text{mV}$ and $V_2 = 103.896\text{mV}$. Therefore, $V_x = V_1 - V_2 = 31.2\text{mV}$.

References

- M. Iordache. Class Lectures, Topic: “Electric circuits.” EEGR 2053, School of Engineering and Engineering Technology, LeTourneau University, 2025.
- M. Iordache. 2024. EEGR 2053 Electric circuits [PowerPoint slides]. Available: <https://mviordache.name/EEGR2053/index.html>
- T. L. Floyd and D. M. Buchla, Principles of Electric Circuits: Conventional Current, 10th ed. Hoboken, NJ: Pearson Education, 2020.
- O. Ortiz. 2025. ENGR 2053 Intro to electric circuits [PowerPoint slides].